
Word Par

Created for Word Enthusiasts and Golf Lovers Alike



Created By: Sleepy Joel (Group #012-1)

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What is Word Par?

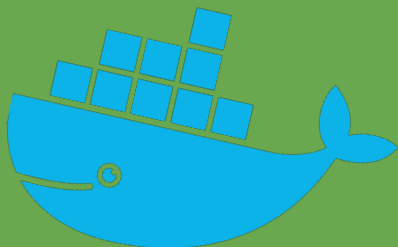
Word Par is a puzzle game that combines golfing and word games to create a unique experience.

- Six Categories: *music, sports, food, animals, nature, and technology*
- Focus on the user experience
- Words are scored semantically and users are provided with the word length
- Competitive aspect through the leaderboard

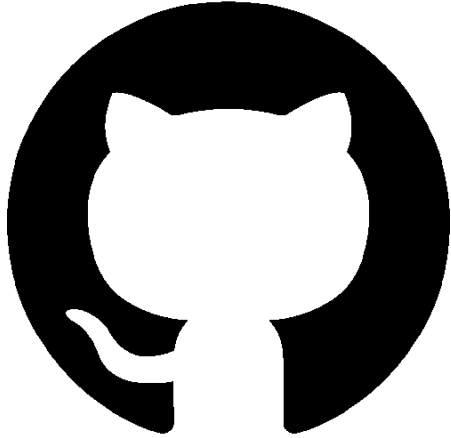
Word Par is a fun, relaxing game the whole family can play!



Tools Used



VCS: GitHub

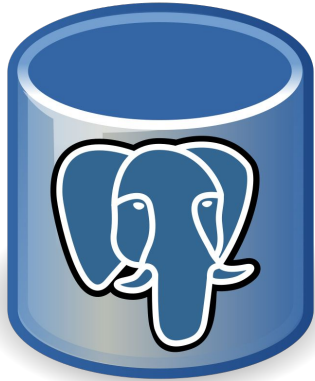


Purpose:

- Supplied a location to host our remote repository
- Equipped our team with a solid foundation for making and tracking changes
- Provided a project board to systematically allocate tasks

Rating: 4.5/5

Database: PostgreSQL

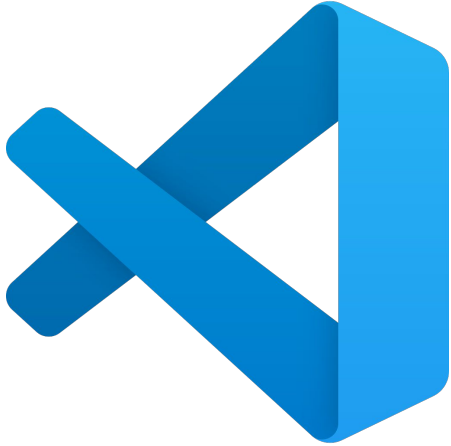


Purpose:

- Relational database management system utilized to power our application's database
- Easy to integrate and use under the umbrella of VSCode IDE, which greatly simplified the building of our database

Rating: 3/5

IDE: VSCode



Purpose:

- Code editor that provides comprehensive tools to make our coding experience more fluent
- Includes Git, diverse libraries, debugging tools, and much more

Rating: 5/5



handlebars



UI Tools: HTML, HandleBars, JS and CSS

Purpose:

- HandleBars was utilized to help organize our HTML structure, by making the HTML dynamic and easier to navigate

Rating: 5/5

Purpose:

- CSS was utilized to help organize the style of our HTML by having a separate style sheet instead of it being built into the HTML code

Rating: 4/5

Purpose:

- JS was utilized to create interactive components in our HTML

Rating: 4.5/5

Application Server: Node.js

Purpose:

- We implemented the Node.js runtime environment in order to help us develop a real-time application on our web server



Rating: 4/5

Deployment Environment: Render

Purpose:



- A simple and unified platform that we employed to host our website and its database
- By using Render, we simplified the process of hosting a website by automating deployment

Rating: 5/5



OpenAI

External API: OpenAI

Purpose:

- Employed for our game to provide users with a word according to the category, and semantically analyze user guesses

Rating: 3.2 /5



OpenAI

External API: OpenAI

How it Works:

- Using OpenAI, there are multiple models to choose from
 - We chose "text-embedding-3-large"
 - Converts text (words, sentences, documents) into numbers
 - Numbers capture the meaning of the text
 - Lets computers compare, search, and analyze text easily
-



OpenAI

External API: OpenAI

Challenges Faced:

- Choosing a model that would meet our requirements
 - Learning each model and how to implement
 - Learning how to connect user requests to an API request through the backend
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OpenAI

External API: OpenAI

Future Scope:

- Implement third-party platforms to provide managed API access
 - Token management system to prevent overuse of the API by certain users
 - Only currently \$5.00 invested in the API
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Testing Tools: Mocha and Chai

Mocha Purpose:

- Testing framework utilized to organize and run tests for applications. Ensures that the code is reliable and efficient

Rating: 5/5

Chai Purpose:

- Assertion library for JavaScript testing. Integrated with the Mocha framework to help verify that the code is behaving as expected

Rating: 4.5/5



Framework: Express.js in Node

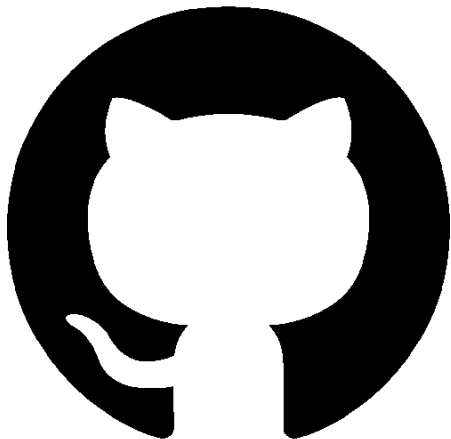
Purpose:

express

- Express.js is used in Node.js to help us create a flexible framework for our web application and our API

Rating: 5/5

Project Tracker: GitHub Projects



Purpose:

- Utilized as a Kanban-style project tracker to view current tasks in the sprint and who it's assigned to
- Highlights what tasks are to be completed in the future and the tasks that we have completed

Rating: 5/5



Methodologies: Agile

Kanban: Through GitHub Projects, we were able to apply a Kanban board to figure out the tasks we had to complete for each sprint, and who should be designated for each user story.

Rating: 4/5

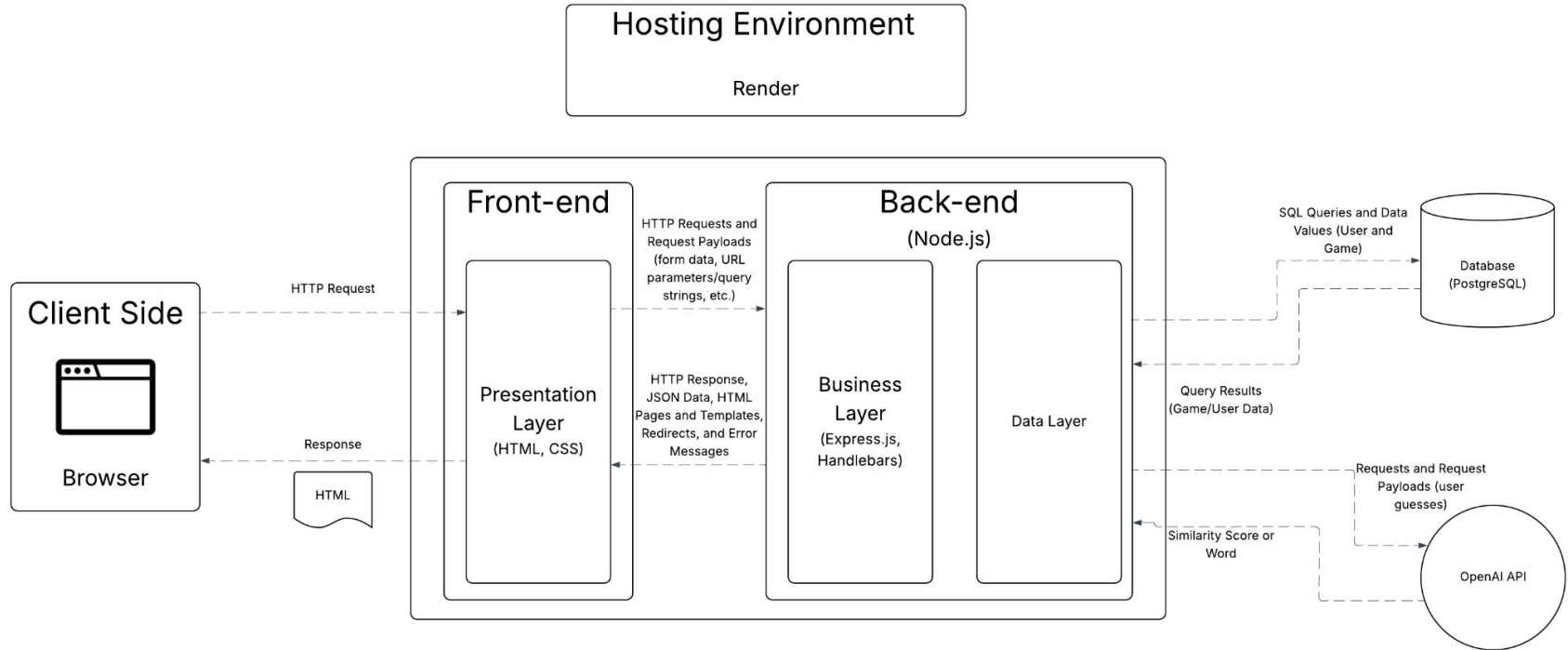
Communication:

- Discord - Utilized different channels to communicate on the different tasks we were completing
 - On Sunday, we discussed how much of the tasks we had assigned were completed and what our plan was for the next couple of days

Meetings:

- TA Meeting Online on Tuesday at 1PM
 - Group Meetings In-Person on Wednesdays at 3:30PM
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Architecture Diagram



Challenges

- **Merge Conflicts**

- Multiple people were working on the same file causing merge conflicts
- To prevent anymore merge conflicts, we collaborated together to ensure we didn't potentially overwrite anyone else's changes

- **API Troubles**

- Many of us had difficulties figuring out how to utilize the API efficiently for our game
- To prevent more API conflicts, we had one specific person focus on the API and learn efficient methods of implementation

- **Debugging CSS & JavaScript**

- Throughout the different pages, combing the JavaScript and CSS into their own files proved difficult to track
 - By adding specific sections throughout their files and utilizing efficient naming conventions, we prevented any overwriting or difficulties
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Future Scope

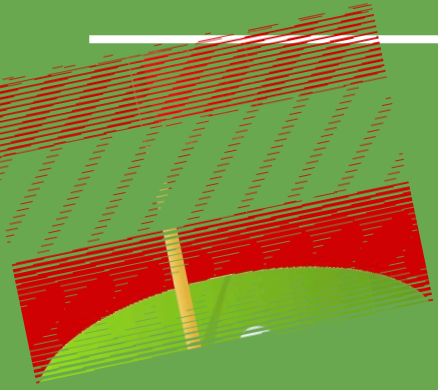
- Adding a difficulty setting
- Providing hints at the cost of score
- Clean up the styling of the game
- Profit making method through a style/skin shop



Demo Video



<https://group-project-wordpar-1.onrender.com/login>



Thank you!
Any Questions?
